



FINAL PROJECT REPORT

**Master of Science (M. Sc) Department
of Tech and Software Major: Data
Science (DS)**

**Analysis of Student Learning
Behavior and Academic
Performance**

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1. Executive Summary

In this report, we present a comprehensive data-analytics study of student learning behaviour and academic performance using Student Performance and Learning Behaviour Dataset obtained from Kaggle. The project investigates the connection between ExamScore and students' learning habits, engagement, motivation, stress, resources, and other characteristics.

The original data set had 14,003 records and 16 attributes. The data-quality checks did not reveal missing values. However, 1,534 exact duplicate rows were found. After excluding these duplicates, there were 12,469 unique records left for the main analysis. The cleaned data was analysed using descriptive statistics, exploratory visualisations, Pearson correlation, ANOVA with Tukey HSD post-hoc testing, and multiple linear regression .

The results suggest that the relationship of individual relationships with ExamScore is weak. The statistically significant effect sizes for StressLevel, Discussions, and AssignmentCompletion are small in practice. The regression model also shows low predictive power ($R^2 \approx 0.006$). The results indicate that the available variables do not explain variation in exam scores very well and should be interpreted as associations and not causal effects.

2. Introduction and Problem Statement

Academic performance can be related to many learning and behavioural factors such as study time, attendance, completion of assignments, motivation, stress, availability of resources and participation in learning activities. Thus, educational data can provide a valuable basis for identifying patterns in student performance, and for identifying which variables merit further investigation.

This project studies these relationships quantitatively based on a structured dataset of student performance. The analysis does not assume that a certain factor leads to better performance, but rather tests whether measurable associations can be found in the available data.

2.1 Problem Statement

Educational institutions gather a significant amount of data concerning their students, but it can be hard to identify the learning and behavioural factors that are associated with academic achievement. In the absence of systematic analysis potentially useful patterns may go undetected.

This project investigates the correlations between study hours, attendance, assignment completion, motivation, stress and other behavioural factors and the academic performance of students by analysing the data on student learning and engagement.

3. Objectives and Research Questions

3.1 Objectives

- Understand and prepare the student performance dataset.
- Analyze student learning and behavioral patterns.
- Examine relationships between learning and engagement factors and ExamScore.
- Investigate study hours, attendance, assignment completion, motivation, and stress in relation to performance.
- Identify factors showing the strongest statistical relationships with academic performance.
- Present findings through statistical analysis, visualizations, and an interactive Power BI dashboard.

3.2 Research Questions

- How does study time relate to students' academic performance?
- What relationship exists between attendance and exam scores?
- Does assignment completion show an association with academic performance?
- How are motivation and stress levels associated with student performance?
- Which learning and behavioral factors show the strongest relationship with academic performance?

4. Dataset Description and Data Preparation

The analysis uses the Student Performance and Learning Behavior Dataset from Kaggle. The dataset contains 14,003 records and 16 attributes representing learning behavior, engagement, personal characteristics, educational technology use, stress, and academic outcomes.

Measure	Value
Original records	14,003
Attributes	16
Missing values	0 across all columns
Exact duplicate rows	1,534
Cleaned records	12,469
ExamScore range	40–100
Average ExamScore	70.31

Table 1. Dataset summary after initial quality assessment.

4.1 Main Variables

The dataset contains StudyHours, Attendance, Resources, Extracurricular, Motivation, Internet, Gender, Age, LearningStyle, OnlineCourses, Discussions, AssignmentCompletion, ExamScore, EduTech, StressLevel, and FinalGrade.

ExamScore is the primary performance outcome in the analysis. FinalGrade is treated as a derived outcome and is excluded from the regression predictors because it is strongly related to ExamScore; including it would risk target leakage and would overstate predictive performance.

4.2 Data Preparation Principles

- Check structure and data types
- Check missing values
- Identify and remove exact duplicate records
- Save a cleaned analytical dataset
- Use the cleaned dataset consistently in downstream analysis

5. System Design and Methodology

The project follows a linear data-analytics pipeline in which each stage feeds the next. The workflow begins with data collection and understanding, continues through cleaning and exploratory analysis, and then applies statistical analysis and regression before communicating results through visualizations and Power BI.

Step	Stage	Details
1	Data collection	Obtain the Kaggle dataset and load it in a Kaggle Notebook.
2	Data understanding	Inspect shape, variables, data types, ranges, and data quality.
3	Data cleaning	Check missing values, detect duplicates, remove exact duplicates, and save the cleaned dataset.
4	EDA	Use descriptive statistics and visualizations to examine distributions and relationships.
5	Correlation analysis	Measure Pearson correlations between numerical variables and ExamScore.
6	Statistical testing	Use ANOVA and Tukey HSD for selected group comparisons.
7	Regression	Fit a multiple linear regression model using an 80/20 train-test split.
8	Visualization & reporting	Build figures, Power BI dashboard, report, and final presentation.

Table 2. Project analytics pipeline.

5.1 Technology Stack

Tool	Purpose
Python	Core analysis language
Pandas / NumPy	Data processing and numerical computation
Matplotlib / Seaborn	Charts and exploratory visualization
SciPy	Statistical testing
Statsmodels	Statistical analysis support
Scikit-learn	Multiple linear regression and evaluation
Kaggle Notebook	Analysis environment
Power BI	Interactive dashboard
GitHub	Version control and project repository
PowerPoint	Final presentation

6. Data Cleaning

The cleaning stage focused on data quality and reproducibility. The original dataset contained no missing values, so no imputation was required. However, 1,534 exact duplicate rows were detected. These duplicates were removed using `pandas drop_duplicates()`, leaving 12,469 unique records for the main analysis.

Check	Original	After cleaning
Rows	14,003	12,469
Missing values	0	0
Duplicate rows	1,534	0

Table 3. Data-quality status before and after duplicate removal.

6.1 Cleaned Dataset

The cleaned dataset was exported as `student_performance_cleaned.csv` and used as the consistent input for subsequent descriptive analysis, correlation analysis, statistical testing, regression, and Power BI. The cleaned file is included in the project repository under `Data/cleaned/`.

6.2 Reproducibility

The cleaning logic is documented in the GitHub `cleaning/data_cleaning.py` script and in the Kaggle Notebook. This separation makes it possible to trace how the analytical dataset was produced without repeating the cleaning process manually.

6.3 Outcome Variable Choice

ExamScore was selected as the main outcome because it directly measures academic performance on a continuous 40–100 scale. FinalGrade is retained for descriptive/dashboard context but is not treated as an independent predictor in the regression model.

7. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was used to understand the overall distribution of ExamScore and to inspect potential relationships with major learning and engagement variables. After cleaning, the average ExamScore was 70.31, with a standard deviation of 17.70 and a range from 40 to 100.

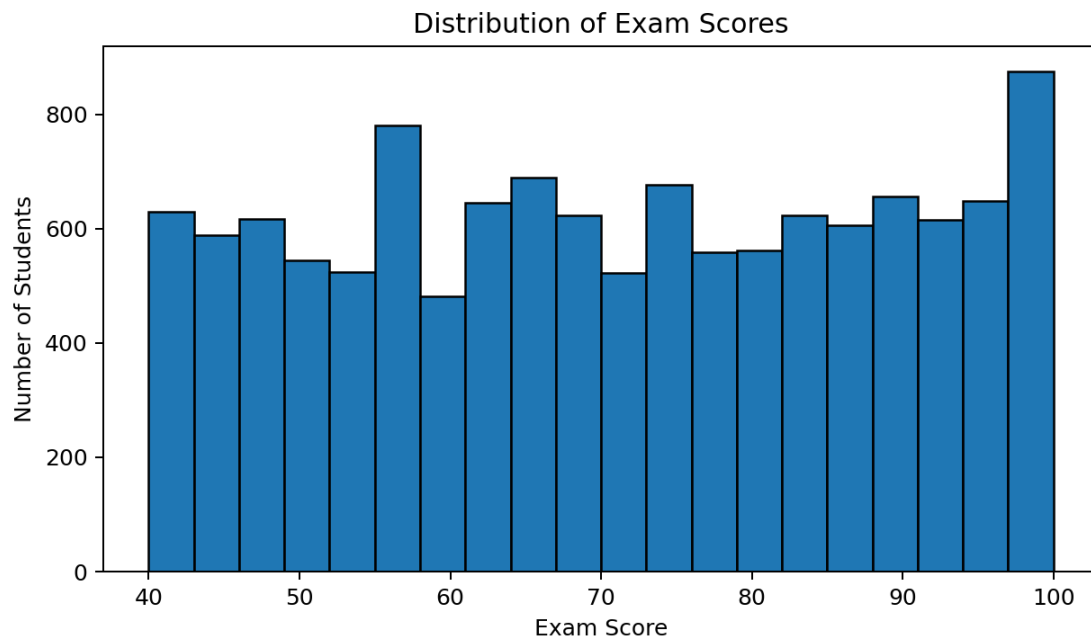


Figure 1. Distribution of ExamScore in the cleaned dataset.

The distribution shows substantial variation in student exam scores. This spread is useful for relationship analysis, but it also means that a predictor must explain meaningful between-student variation to be useful in a predictive model.

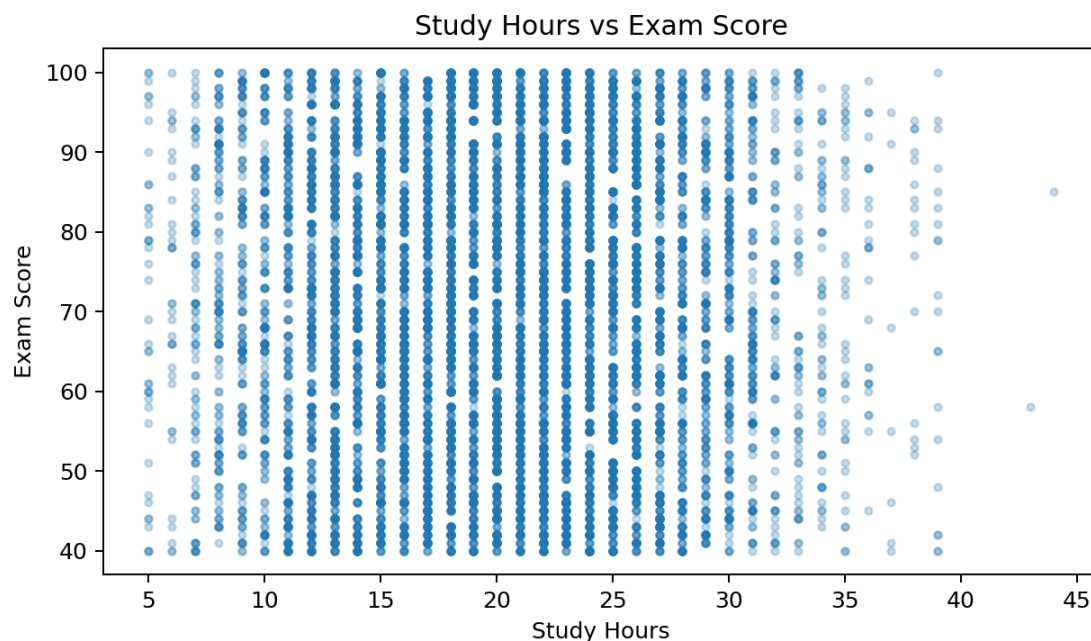


Figure 2. StudyHours versus ExamScore. The visual relationship is weak.

The scatter plot does not show a clear linear trend between StudyHours and ExamScore. This visual observation is consistent with the near-zero Pearson correlation obtained later in the analysis.

7. Exploratory Data Analysis (continued)

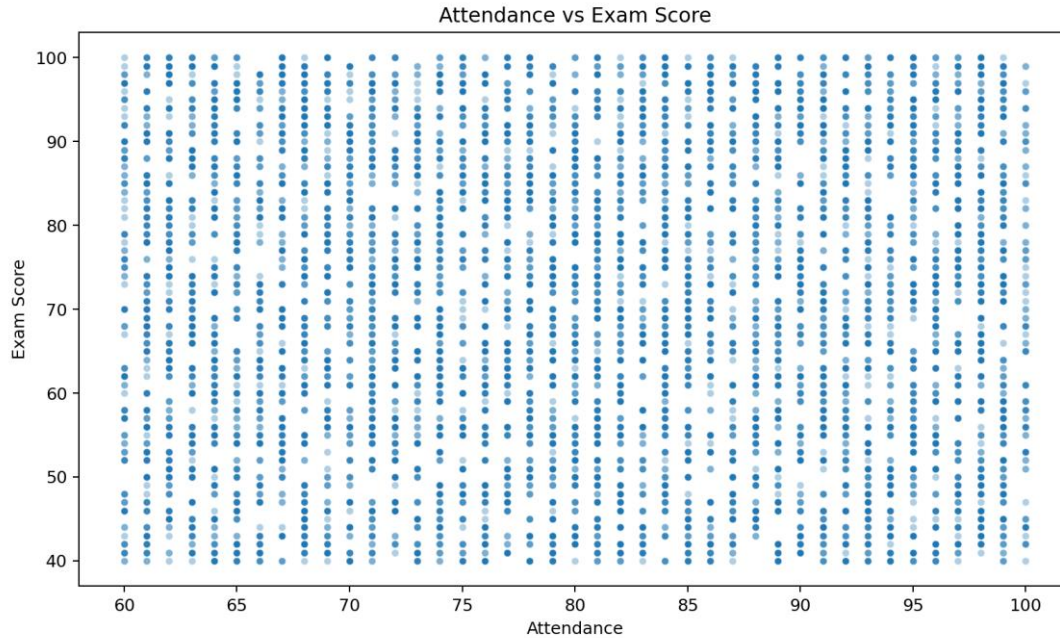


Figure 3. Attendance versus ExamScore.

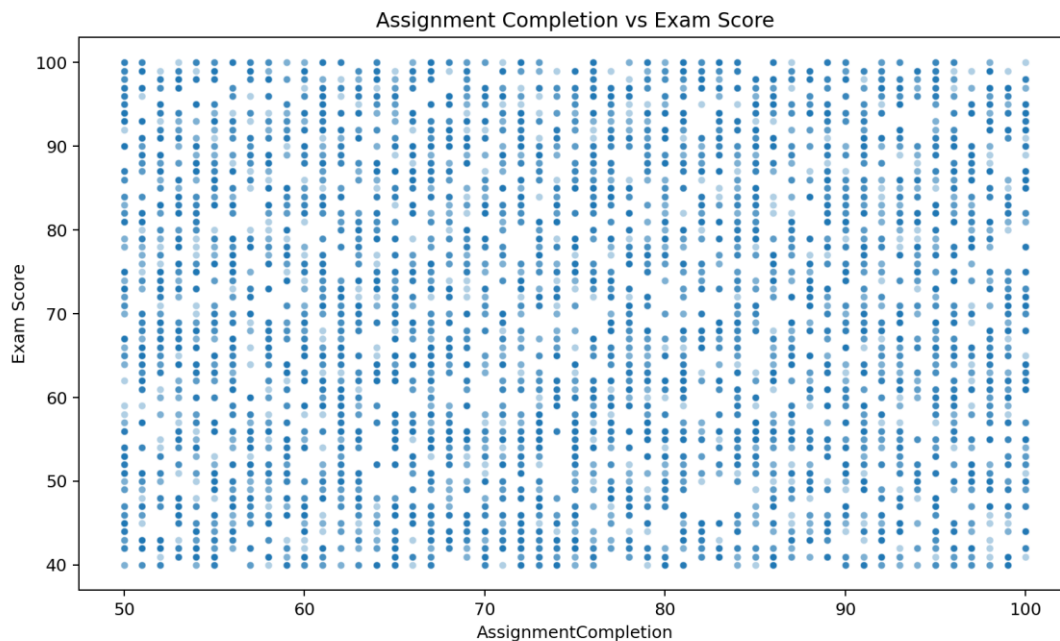


Figure 4. AssignmentCompletion versus ExamScore.

Both scatter plots show considerable dispersion. Attendance does not exhibit a strong linear pattern, while AssignmentCompletion shows a small positive association that becomes statistically detectable because the dataset contains many observations.

7.1 Group Comparisons

Average exam scores were also compared across selected categorical variables. Motivation groups have very similar means, whereas stress groups show more visible differences. These group-level patterns are examined formally with ANOVA in Section 9.

8. Correlation Analysis

Pearson correlation was used to quantify linear relationships between numerical variables and ExamScore. The analysis excluded FinalGrade from predictor interpretation because it is a derived outcome closely associated with ExamScore.

Variable	Pearson r	p-value
StressLevel	-0.0343	0.000126
Discussions	-0.0312	0.000491
AssignmentCompletion	0.0274	0.002212
OnlineCourses	0.0207	0.020904
Gender	0.0167	0.061442
Attendance	-0.0142	0.113955
Age	-0.0101	0.257682
EduTech	0.0103	0.249597
LearningStyle	0.0083	0.355096
Extracurricular	-0.0068	0.446316
Internet	-0.0073	0.412353
Resources	0.0051	0.567600
StudyHours	0.0042	0.639007
Motivation	-0.0035	0.692705

Table 5. Pearson correlations with ExamScore.

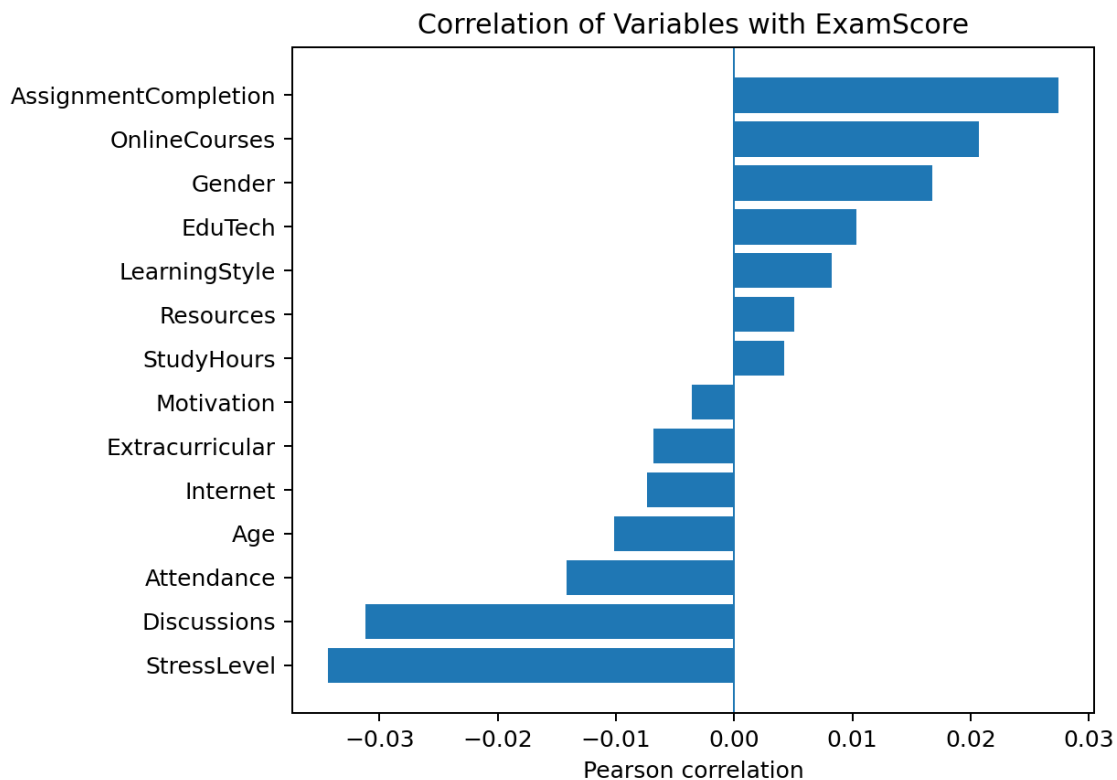


Figure 5. Correlation of numerical variables with ExamScore.

The largest absolute correlations are still small. StressLevel, Discussions, and AssignmentCompletion meet the conventional 0.05 significance threshold, but the magnitudes indicate limited practical association. OnlineCourses is statistically detectable at the 0.05 level but does not survive a stricter Bonferroni-adjusted threshold.

9. Statistical Analysis

Statistical tests were used to move beyond descriptive comparisons. One-way ANOVA was applied to compare mean ExamScore across Motivation and StressLevel groups. Tukey HSD was used after the significant stress ANOVA to identify which group pairs differed.

Factor	F-statistic	p-value	Conclusion
StressLevel	12.9840	< 0.001	Significant group difference
Motivation	0.9058	0.4043	Not significant

Table 6. One-way ANOVA results.

9.1 StressLevel ANOVA and Tukey HSD

StressLevel produced a statistically significant ANOVA result ($F = 12.9840$, $p < 0.001$). The eta-squared effect size is approximately 0.0021, which is very small. Tukey HSD indicates that StressLevel 0 differs significantly from levels 1 and 2, while levels 1 and 2 are not significantly different from each other.

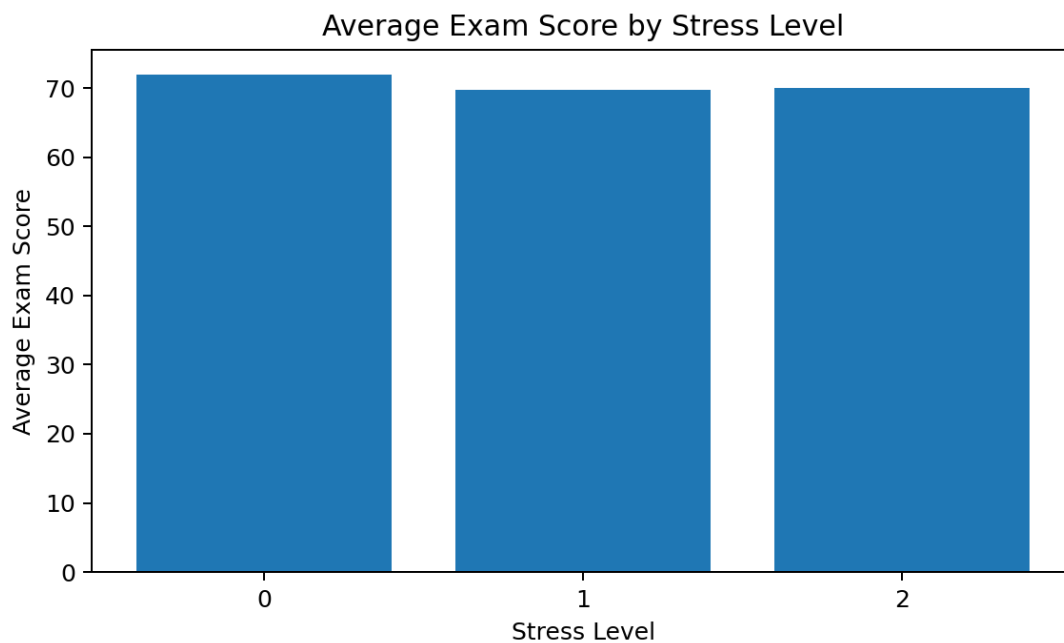


Figure 6. Average ExamScore by StressLevel.

9.2 Interpretation

The statistical significance is influenced by the large sample size, whereas the effect size shows that stress category explains only a very small fraction of score variation. Therefore, the result is statistically meaningful but not strong enough to support a large practical effect.

10. Multiple Linear Regression

A multiple linear regression model was fitted to assess the combined relationship between the available learning and behavioral variables and ExamScore. FinalGrade was excluded to avoid leakage from a derived outcome. The data were split into 80% training and 20% testing observations using `random_state = 42`.

Metric	Value
R^2	0.0060
RMSE	17.69
Train/test split	80% / 20%
Random state	42

Table 7. Regression model evaluation.

An R^2 of approximately 0.006 means that the model explains only about 0.6% of the variation in ExamScore in the test data. The RMSE of about 17.69 points is large relative to the observed score range. The model therefore has weak predictive performance.

10.1 Selected Coefficients

Feature	Coefficient
Discussions	-1.019
Internet	-0.673
StressLevel	-0.638
Gender	0.580
EduTech	0.144
LearningStyle	0.082
StudyHours	0.037
AssignmentCompletion	0.035

Table 8. Selected regression coefficients; coefficients are conditional associations in the fitted model.

11. Power BI Dashboard

An interactive Power BI dashboard was developed to communicate the main findings visually. The dashboard combines KPI cards, scatter plots, categorical comparisons, and slicers for Motivation and StressLevel.



Figure 8. Power BI dashboard used for interactive project presentation.

11.1 Dashboard Components

- 12,469 records represented by the cleaned analytical dataset
- Average ExamScore, StudyHours, Attendance, and AssignmentCompletion KPIs
- StudyHours, Attendance, and AssignmentCompletion versus ExamScore scatter plots
- Average ExamScore by Motivation, StressLevel, Discussions, OnlineCourses, and LearningStyle
- FinalGrade views for descriptive context
- Interactive Motivation and StressLevel slicers

The dashboard is intended as a communication layer rather than a replacement for the statistical analysis. The Python notebook remains the main reproducible analysis record, while Power BI provides an accessible interactive view for presentation and exploration.

12. Key Findings and Discussion

- The cleaned analytical dataset contains 12,469 unique records after removing 1,534 exact duplicates.
- StudyHours and Attendance show very weak linear relationships with ExamScore.
- AssignmentCompletion, Discussions, and StressLevel show statistically significant but small associations with ExamScore.
- StressLevel has a statistically significant group difference, but the effect size is very small.
- The regression model has $R^2 \approx 0.006$, indicating weak overall predictive/explanatory power.
- The results should be interpreted as associations rather than evidence of causation.

13. Limitations, Conclusion and Project Deliverables

13.1 Limitations

- The analysis is limited to the variables available in the Kaggle dataset.
- The study is observational and does not establish causal relationships.
- Most observed relationships are weak, which limits predictive usefulness.
- FinalGrade is strongly related to ExamScore and was excluded from regression predictors to reduce leakage.
- The project does not model longitudinal student trajectories because the available data are not structured as repeated observations over time.

13.2 Conclusion

This project demonstrates a complete and reproducible data-analytics workflow for studying student learning behavior and academic performance. The analysis shows that several behavioral variables have statistically detectable associations with ExamScore, but the observed effects are small and the overall predictive model is weak. The most important conclusion is therefore not that one factor strongly determines performance, but that academic outcomes in this dataset are not well explained by the available variables considered individually or together.

The combination of statistical analysis, visualization, GitHub documentation, and the Power BI dashboard provides a consistent evidence-based presentation of the project results. Future work could incorporate richer behavioral, demographic, longitudinal, or contextual variables and evaluate more flexible predictive methods.

13.3 Project Deliverables and Repository

- Cleaned dataset
- Kaggle analysis notebook
- Analysis scripts for cleaning, EDA, correlation, statistical analysis, and regression
- Power BI dashboard evidence
- Project proposal, mindmap, and Gantt chart
- Final project report and presentation

Repository: https://github.com/Aiman22-33/57112524_Datascience120A

References

- Kaggle. Student Performance and Learning Behavior Dataset. Primary dataset used for the project.
- Project repository: https://github.com/Aiman22-33/57112524_Datascience120A
- Python libraries used: Pandas, NumPy, Matplotlib, Seaborn, SciPy, Statsmodels, and Scikit-learn.